

Introduction to AI Ethics Ethics, Compliance, and Core Principles for Engineers

Course: Ethics and Risk Mitigation in AI
UCS421

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Why Talk About Ethics in AI?

- AI systems influence real-world decisions
- Decisions affect:
 - Safety
 - Access to services
 - Rights and dignity
 - Public trust
- Failures can cause harm at scale

How AI Is Different from Traditional Software

- Learns from data, not fixed rules
- Probabilistic behavior
- Changes after deployment
- Hard to fully predict or explain
- Ethics becomes critical under uncertainty

Technical Failure vs Ethical Failure

Technical Failure

- Bug
- Sensor malfunction
- Wrong parameter

Ethical Failure

- Discrimination
- Unfair risk exposure
- Privacy violation
- Lack of accountability

What Is Compliance?

Compliance means:

- Following laws and regulations
- Meeting standards and policies
- Passing audits and certifications

Key question: “Are we following the rules?”

What Is Ethics?

Ethics means:

- Making responsible judgments
- Anticipating harm
- Considering societal impact
- Acting beyond minimum requirements

Key question: “Should we do this, even if it’s allowed?”

Ethics vs Compliance

COMPLIANCE	ETHICS
Rule-based	Principle-based
Mandatory	Reflective
Reactive	Proactive
Minimum standard	Higher responsibility

Why Compliance Is Not Enough

- Laws lag behind technology
- Regulations don't cover all edge cases
- Legal ≠ ethical
- Harm can occur without violations

Core Ethical Principles in AI

- Fairness
- Accountability
- Transparency
- Privacy

These guide responsible AI design and deployment

What Is Fairness in AI?

Fairness means:

- Avoiding systematic disadvantage
- Ensuring consistent system performance
- Preventing unequal harm across users or groups

Fairness is **context-dependent**

Sources of Unfairness

- Biased or incomplete data
- Measurement or sensor errors
- Historical inequalities in data
- Design choices favoring averages
- Ignoring edge cases

Fairness in Engineering Systems

Examples:

- Vision systems failing in low light
- Predictive models underestimating rare failures
- Access systems misidentifying certain users

Unfairness can become a **safety risk**

Fairness Is a Design Responsibility

Engineers must:

- Test across conditions
- Examine subgroup performance
- Justify acceptable trade-offs
- Anticipate who may be harmed

Use Case 1: Campus Face Recognition System

A university installs an AI-based face recognition system to allow students entry into hostels and labs. The system works very well during the day but frequently fails at night or in poor lighting. Some students are repeatedly denied entry, while others rarely face problems.

Discussion prompts:

- Is this system fair if it works well on average?
- Should students who face more failures bear the inconvenience?
- Who should adjust—the students or the system?

Principle illustrated:

Fairness is about **consistent performance across users and conditions**, not just overall accuracy.

Use Case 2: AI-Based Scholarship Shortlisting

An AI system shortlists students for a merit-based scholarship using past academic records. Students from certain schools are consistently shortlisted more often because historical data reflects better resources in those schools.

Discussion prompts:

Is the AI system being unfair, or just “data-driven”?

Should engineers intervene if the outcome disadvantages some groups?

What would fairness mean in this context?

Principle illustrated:

Fairness requires questioning **what the data represents**, not just how well the model predicts.

What Is Accountability?

Accountability means:

- Knowing who is responsible
- Being able to explain decisions
- Taking corrective action after harm

Key question: “Who answers when the system fails?”

Accountability Challenges in AI

- Automated decision-making
- Multiple teams involved
- Black-box models
- Responsibility diffusion

Result: **Accountability gap**

Engineering for Accountability

Engineers enable accountability through:

- Documentation
- Traceability
- Logging and monitoring
- Clear human oversight
- Audit-ready systems

Use Case 3: AI Attendance System

A company uses an AI-based camera system to mark employee attendance. One day, the system incorrectly marks several employees absent, affecting their pay. Management says, “The system did it, not us.”

Discussion prompts:

Who should be held responsible for the error?

Can responsibility be assigned to an AI system?

What should accountability look like here?

Principle illustrated:

Accountability means **humans remain responsible**, even when decisions are automated.

Use Case 4: Smart Traffic Signal

An AI-controlled traffic signal prioritizes traffic flow but fails to give enough crossing time for elderly pedestrians. An accident occurs, and multiple agencies are involved in the system's design.

Discussion prompts:

Who is accountable: engineers, city authorities, or vendors?

Should responsibility be shared?

How could accountability have been clearer?

Principle illustrated:

Without clear accountability structures, responsibility becomes diffused and unclear.

What Is Transparency?

Transparency means:

- System behavior is understandable
- Decisions can be explained
- Assumptions and limits are visible

Key question: “Can humans question the system?”

Why Transparency Matters

Transparency supports:

- Trust
- Debugging
- Auditing
- Oversight
- Safe human-AI interaction

Opaque systems = hidden risks

Transparency $\neq$ Full Disclosure

Transparency does NOT require:

- Revealing source code
- Explaining every equation

Transparency DOES require:

- Meaningful explanations
- Clear limitations
- Honest communication

Use Case 5: AI Exam Evaluation

An AI system grades short-answer exams. Some students receive unexpectedly low marks. When they ask for clarification, the only response is: “The AI evaluated your answer.”

Discussion prompts:

Is this acceptable?

What kind of explanation should students receive?

Does transparency mean showing the algorithm?

Principle illustrated:

Transparency requires **meaningful explanations**, not technical details.

Use Case 6: AI Health Risk App

A health app tells users they are “high risk” but does not explain why. Some users panic; others ignore the warning.

Discussion prompts:

How does lack of explanation affect trust?

What minimum transparency is needed in health-related AI?

Who needs the explanation—doctors, users, or both?

Principle illustrated:

Transparency supports trust and appropriate human response.

What Is Privacy?

Privacy concerns:

- Data collection
- Data use
- Data storage
- Data sharing

Key question: “Is this data use justified and proportionate?”

Why Privacy Is Critical in AI

AI systems often:

- Collect large volumes of data
- Combine multiple data sources
- Infer sensitive attributes
- Retain data long-term

Privacy Risks in Engineering AI

Examples:

- Smart sensors revealing personal habits
- Health monitoring exposing conditions
- Location data enabling surveillance
- Industrial data leaks

Privacy-by-Design

Ethical AI requires:

- Data minimization
- Purpose limitation
- Secure storage
- Access control
- Defined retention periods

Use Case 7: Smart Classroom Monitoring

A smart classroom system tracks student attention, attendance, and movement to improve learning outcomes. Students are not told exactly what data is collected or how long it is stored.

Discussion prompts:

Is this a privacy violation?

Should consent be required?

Where is the line between improvement and surveillance?

Principle illustrated:

Privacy concerns **control and awareness**, not just security.

Use Case 8: Fitness App Data Sharing

A fitness app collects step count and heart rate data. Later, users discover that anonymized data was shared with third parties for research.

Discussion prompts:

Is anonymization enough to protect privacy?

Should users have been informed?

Can good intentions still lead to ethical problems?

Principle illustrated:

Privacy includes **purpose limitation and user expectations**, not just data protection.

Ethical Principles Are Connected

- Transparency → Accountability
- Fairness → Needs transparency
- Privacy → Affects data and performance
- Trade-offs are unavoidable

Engineers as Ethical Decision-Makers

- Design choices embed values
- Engineers allocate risk
- Ethical responsibility starts at design
- Cannot be outsourced

Key Takeaways

- Ethics goes beyond compliance
- Fairness, accountability, transparency, and privacy are core
- Ethical failures can occur without bugs
- Responsible AI is good engineering